

Version 11 Integration Supplement: Boundary-Complete Layer B, CDT Reconstruction, Layer-C No-Go, and Finite-Place Functoriality

Apéry-systems programme

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Abstract

This supplement records three upgrades to the three-layer Hecke–Eisenstein/Apéry classification. First, the previously open boundary completion of Franke’s critical-value kernel is replaced by an exact modular theorem: in weights at least three Franke’s local weak-function residue term vanishes automatically, and the missing Layer-B data are supplied instead by principal-character Eisenstein boundary directions and a finite cusp-constant matrix. Second, the successful Calegari–Dimitrov–Tang proof of the $\mathbf{Q}$ -linear independence of $1, \zeta(2), L(2, \chi_{-3})$ is reconstructed inside the adelic Layer-C formalism. Their exact Eisenstein inputs are the inner/outer orientations predicted by the classification; their fourteen-function denominator array becomes the free-depth flag $14 \rightarrow 3 \rightarrow 1$ and reproduces the flat penalty $191/49$ exactly. The comparison also forces one further Layer-C invariant, the horizontal-integration profile e , whose additional cost is $27/80$. Thus the complete successful arithmetic cost is $16603/3920$. Third, the $L(3, \chi_5)$ six-sheet programme yields a new monotone entry-obstruction principle: a curvature- ≤ -1 submetric on a convenient hyperbolic super-surface can rule out an adelic realization before the full CDT energy is computed. An explicit theta/Fourier submetric gives a robust computer-assisted level-20 no-go certificate.

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1 Revised three-layer datum

The classification object is

$$(\mathbb{V}, [\nu], \mathcal{R}_{\text{ad}}).$$

Layers A and B are unchanged in substance: $\mathbb{V}$ is the pure/direct-image local system and $[\nu]$ is its mixed Eisenstein extension together with its critical, orientation and boundary coordinates. The CDT calibration requires one refinement of Layer C.

Definition 1.1 (Fine adelic Apéry realization). *A fine adelic realization of $(\mathbb{V}, [\nu])$ consists of*

$$\mathcal{R}_{\text{ad}} = (\sigma_{\text{dR}}, a, (t, \varphi), \Lambda, U^\bullet, e, \{\Phi_p, \kappa_p\}_{p \notin S}).$$

Here:

- (i) σ_{dR} is the marked cyclic/de Rham section;
- (ii) a is scalar or vector companion descent;
- (iii) (t, φ) consists of the arithmetic coordinate and a common complex continuation pullback;
- (iv) Λ is the integral lattice;
- (v) $U^\bullet$ is a nested free-depth filtration controlling LCM columns;
- (vi) e is the horizontal-integration profile recording controlled integrations not already represented by the LCM columns;
- (vii) (Φ_p, κ_p) records the finite-place homogeneous Frobenius and extension coordinate wherever available.

The extra datum e is forced by the successful CDT benchmark. It is logically independent of the free-depth filtration: differentiation, algebraic descent and horizontal integration can change the fine arithmetic-holonomy cost without changing the underlying Layer-A local system or Layer-B regulator class.

Definition 1.2 (Fine arithmetic penalty). *Let an m -dimensional holonomy tuple have LCM columns of scales b_j and let the j th column vanish on an initial adapted subspace of dimension u_j . Define*

$$\tau_{\text{F}} = \sum_j b_j \left(1 - \frac{u_j^2}{m^2}\right).$$

If the realization also carries a CDT horizontal-integration profile e , denote the corresponding fine integration cost by $\tau_{\text{hor}}(e)$ and set

$$\tau_{\text{ad}} = \tau_{\text{F}} + \tau_{\text{hor}}(e).$$

The resulting fine adelic score is

$$\text{Score}_{\text{ad}} = m(\log |\varphi'(0)| - \tau_{\text{ad}}) - E(\varphi).$$

The first summand is the Frobenius/free-depth reinterpretation of the CDT denominator-array penalty; the second is not inferred from Frobenius data and must be computed from the actual horizontal integration pattern. In particular, a claimed adelic comparison with CDT is incomplete unless both terms are matched.

2 Boundary-complete Franke Layer B

The earlier foundational draft left the passage from Franke’s nonprincipal critical kernel to the principal-character regulator sector as an open “boundary-completed Franke sequence.” The completed calculation changes both the statement and the mechanism.

Proposition 2.1 (Automatic disappearance of the weak-function residue). *Let $k \geq 3$ and let ω, η be weak functions in Franke’s sense, with poles of order at most one at rational points. Then*

$$\text{Res}_{z=0}(z^{k-1}\eta(z)\widehat{\omega}(z/\tau)) = 0.$$

Thus the local residue term in the weak-function transformation is not the missing boundary coordinate in the weights relevant to the present theory.

Proof. The product of the two weak functions has pole order at most two. Multiplication by z^{k-1} leaves lowest exponent $k-3 \geq 0$, so no z^{-1} term remains. $\square$

Let F denote a Franke nonprincipal Eisenstein sector and let

$$K_F = \ker \mathcal{L}_S$$

be its exact prescribed-critical-value kernel. Adjoin the two primitive principal-character orientations

$$S_\chi = E_{k,\chi,1}, \quad T_\chi = E_{k,1,\bar{\chi}}.$$

Their constant terms at the two distinguished cusps are transverse: one is nonzero only at 0, the other only at ∞ (up to the usual Fricke normalization). Put $P_\chi = \text{Span}\{S_\chi, T_\chi\}$ and $E = F \oplus P_\chi$.

Theorem 2.2 (Split two-cusp boundary completion). *For any linear regularized extension $\mathcal{L}_S^{\text{reg}}$ of Franke’s critical map to E , the sequence*

$$0 \longrightarrow K_F \longrightarrow E \xrightarrow{(\mathcal{L}_S^{\text{reg}}, \partial_{0,\infty})} C_{\text{crit}} \oplus \mathbf{C}^2 \longrightarrow 0$$

is exact. Its kernel is independent of the chosen endpoint regularization.

Proof. The boundary map is an isomorphism on P_χ , while every element of F vanishes at the two distinguished cusps. Hence the boundary coordinates recover the principal-character component uniquely. Franke’s critical map is surjective on F , which gives surjectivity of the combined map. Vanishing boundary forces the P_χ component to vanish; the remaining critical equation is precisely membership in K_F . Two regularizations differ by a linear function of the two boundary coordinates, hence by a shear of the target and have the same kernel. $\square$

Now let ∂_* denote all remaining cusp constant terms. Set

$$K_\partial = K_F \cap \ker \partial_*.$$

Theorem 2.3 (Full-cusp Layer-B kernel). *The full boundary-completed kernel is computed by the finite cusp matrix*

$$\partial_*|_{K_F} : K_F \rightarrow B_*.$$

In particular,

$$\boxed{\dim K_\partial = \dim K_F - \text{rank}(\partial_*|_{K_F}).}$$

After decomposition under any semisimple normalizer/diamond action preserving the data, the same statement holds sector by sector and yields the canonical filtration

$$\boxed{K_\partial \subseteq K_{\text{or}} \subseteq K_{\text{sc}}.}$$

The level-100 Franke- η calculation gives the sharp worked example

$$\dim K_{\text{sc}} = 3, \quad \dim K_{\text{or}} = 1, \quad \dim K_{\partial} = 0.$$

Thus critical invisibility, orientation invisibility and boundary invisibility are genuinely different Layer-B notions. The formerly open “boundary completion” is therefore reduced to exact finite Bernoulli/cusp linear algebra once a Franke critical-kernel basis is known.

3 The CDT $L(2, \chi_{-3})$ proof as an adelic Layer-C theorem

Put

$$\Lambda = L(2, \chi_{-3}), \quad S = E_{3, \chi_{-3}, 1}, \quad T = E_{3, 1, \chi_{-3}}.$$

The exact source comparison established in the programme is

$$\boxed{D^2 B_{\text{CDT}} = (1 - 8V_2)S, \quad D^2 C_{\text{CDT}} = (1 - V_2)T.} \quad (1)$$

The corresponding connection constants are

$$\frac{1}{2}\Lambda, \quad \frac{1}{4}\zeta(2).$$

Hence the successful CDT proof uses exactly the two primitive Eisenstein orientations predicted by Layer B, but does not fully purify away the Tate direction.

3.1 Target-zero extension under a rational relation

Let A be the common weight-one elliptic period on the level-6 host and write

$$H_A = A, \quad H_B/H_A = B_{\text{CDT}}, \quad H_C/H_A = C_{\text{CDT}}.$$

CDT prove integral/LCM arithmetic for the corresponding Taylor coefficients and prove that

$$H_B - \frac{\Lambda}{2}H_A, \quad H_C - \frac{\zeta(2)}{4}H_A$$

overconverge through the nearest singularity.

If

$$a + b\frac{\Lambda}{2} + c\frac{\zeta(2)}{4} = 0$$

with $a, b, c \in \mathbf{Q}$, then

$$H = aH_A + bH_B + cH_C = b\left(H_B - \frac{\Lambda}{2}H_A\right) + c\left(H_C - \frac{\zeta(2)}{4}H_A\right)$$

is a rational target-zero realization. Thus the contradiction hypothesis produces exactly the affine Layer-B phenomenon predicted by the classification.

3.2 Normalizer quotient and fourteen-dimensional holonomy

CDT use the involution

$$w(x) = \frac{x}{x-1}$$

and the invariant coordinate

$$y = x + w(x) = \frac{x^2}{x-1}.$$

The plus symmetrization $G(y) = H(x) + H(w(x))$ passes from one branch of the conditional extension to a normalizer quotient. CDT then prove that seven functions generated from G by differentiation and controlled integration, together with seven unconditional “pure” functions, are linearly independent over $\mathbf{Q}(y)$. Hence the contradiction hypothesis produces a holonomy tuple of exact dimension

$$\boxed{m = 14.}$$

3.3 The free-depth flag reproduces CDT's flat penalty exactly

In a basis adapted to CDT's denominator array there are two LCM layers, both of scale 2. The first is absent on a three-dimensional initial subspace and the second on a one-dimensional initial subspace. Equivalently the Layer-C filtration is

$$\boxed{V = U^0 \supset U^1 \supset U^2, \quad \dim(U^0, U^1, U^2) = (14, 3, 1).}$$

Therefore Definition 1.2 gives

$$\begin{aligned} \tau_F &= 2 \left(1 - \frac{3^2}{14^2} \right) + 2 \left(1 - \frac{1^2}{14^2} \right) \\ &= \boxed{\frac{191}{49}}. \end{aligned}$$

This is exactly the “flat” denominator penalty appearing in the CDT proof. Thus the free-depth interpretation is calibrated on a theorem that actually proves irrationality; it is not merely a heuristic rephrasing.

3.4 Horizontal integrations force one additional Layer-C datum

The successful fourteen-function basis also contains controlled horizontal integrations not represented by the two LCM columns. In CDT's ordered basis the integration profile is

$$\boxed{e = (0, 0, 1; 0, 0, 0, 0, 0, 0; 1, 1, 1, 1, 1).}$$

Their fine holonomy theorem assigns the additional cost

$$\boxed{\tau_{\text{hor}}(e) = \frac{27}{80}}.$$

Consequently the complete arithmetic cost is

$$\boxed{\tau_{\text{ad}} = \frac{191}{49} + \frac{27}{80} = \frac{16603}{3920}}. \tag{2}$$

This is the reason the horizontal-integration profile is now part of Definition 1.1.

Theorem 3.1 (CDT reconstruction theorem). *The Calegari–Dimitrov–Tang proof of the $\mathbf{Q}$ -linear independence of*

$$1, \quad \zeta(2), \quad L(2, \chi_{-3})$$

can be organized exactly as a three-layer Hecke–Eisenstein/adelic realization with the following data:

- (i) *Layer A is the common rank-two elliptic local system generated by the weight-one period A;*
- (ii) *Layer B consists of the two orientations in (1), with regulator coordinates $\Lambda/2$ and $\zeta(2)/4$;*
- (iii) *a rational relation among $1, \Lambda, \zeta(2)$ creates a rational target-zero extension;*
- (iv) *normalizer descent produces a fourteen-dimensional common holonomy tuple;*
- (v) *the exact free-depth flag is $14 \rightarrow 3 \rightarrow 1$ and gives $\tau_F = 191/49$;*
- (vi) *the horizontal-integration profile contributes $\tau_{\text{hor}} = 27/80$;*

(vii) therefore the fine adelic penalty is $\tau_{\text{ad}} = 16603/3920$;

(viii) CDT's quantitative holonomy inequality gives an upper bound strictly below 14 for this tuple, contradicting the proved functional independence.

Hence the adelic Layer-C formalism reconstructs the complete architecture of a successful irrationality/linear-independence proof for a non-Tate Dirichlet L -value.

Proof. Items (i)–(iii) follow from the source identities, the two connection constants, and the target-zero rewriting above. Item (iv) is CDT's normalizer descent and functional-independence theorem. Item (v) is the direct evaluation of their denominator array in the free-depth formula. Item (vi) is their fine integration penalty; summing gives (2). Finally CDT construct a common analytic pullback and evaluate the fine arithmetic-holonomy bound, obtaining a dimension bound smaller than the fourteen independent functions, which contradicts the assumed rational period relation. The conclusion is exactly their linear-independence theorem, now expressed in the three-layer dictionary. $\square$

Corollary 3.2 (Task-dependent purification is Diophantine, not cosmetic). *Full Layer-B purification is not universally optimal for Layer-C Diophantine arguments. In the CDT example the partially purified inner and outer directions form a more useful arithmetic-holonomy tuple than the fully inner-purified scalar source*

$$(1 - 2V_2)(1 - 8V_2)S.$$

Remark 3.3 (What has and has not been calibrated). *The successful comparison calibrates the critical/orientation source dictionary, target-zero affine extension, normalizer descent, free-depth filtration and horizontal-integration penalty. It does not by itself identify the local Frobenius extension coordinate κ_p with a normalized p -adic Dirichlet L -value. That syntomic identification remains a separate open theorem.*

4 A monotone entry obstruction for adelic Layer C

The arithmetic-holonomy theorem has an entry hypothesis before its full energy bound becomes relevant: a common analytic pullback must have enough local conformal radius to overcome the arithmetic penalty. This yields a useful no-go principle which is cheaper than computing the complete hyperbolic metric or the CDT energy.

Theorem 4.1 (Monotone entry obstruction). *Let X be the continuation surface of an adelic Apéry realization, let $z_0 \in X$ be the Taylor point, let x be the normalized arithmetic coordinate, and let τ_{ad} be any proved lower bound for the arithmetic-holonomy penalty of the chosen holonomy tuple. Suppose $X \subset Y$ with Y hyperbolic and suppose $\sigma(z)|dz|$ is a conformal pseudometric on Y of curvature at most -1 . Then*

$$\rho_X(z_0) \geq \rho_Y(z_0) \geq \sigma(z_0),$$

and therefore

$$r_x(X, z_0) \leq \frac{2|x'(z_0)|}{\sigma(z_0)}. \quad (3)$$

In particular, if

$$\frac{2|x'(z_0)|}{\sigma(z_0)} \leq e^{\tau_{\text{ad}}}, \quad (4)$$

then the CDT entry condition $\log r_x > \tau_{\text{ad}}$ is impossible and no full energy calculation is needed for that realization.

Proof. Domain monotonicity gives $\rho_X \geq \rho_Y$. Ahlfors–Schwarz gives $\sigma \leq \rho_Y$. With curvature- -1 disc normalization $\rho_{\mathbb{D}}(0) = 2$, the conformal radius in the local coordinate x is $r_x = 2|x'(z_0)|/\rho_X(z_0)$. Combining the inequalities gives (3); (4) contradicts the entry hypothesis. $\square$

This theorem belongs to Layer C rather than Layer B: it can rule out a Diophantine realization even when the local system and regulator extension have been classified and purified perfectly.

5 The $L(3, \chi_5)$ level-10 entry obstruction

The first fold-pure direction on the level-10 genus-zero cover is

$$G_{10} = 3S + 8V_2S.$$

It removes the first conjugate Tate fold pair but leaves the second pair at

$$u_{\pm} = \frac{-2 \pm i}{10}.$$

Even if every other singularity is filled, the continuation surface is contained in

$$\mathbb{C} \setminus \{u_+, u_-\}.$$

After affine normalization to $\mathbb{C} \setminus \{0, 1\}$ the Taylor point becomes $z_0 = 1/2 - i$. Let $\lambda(\tau_0) = z_0$. The inverse-lambda formula

$$\tau_0 = i \frac{K(1 - z_0)}{K(z_0)}$$

and complex AGM iteration give the coarse enclosure

$$-0.533 < \Re \tau_0 < -0.532, \quad 0.846 < \Im \tau_0 < 0.847.$$

The twice-punctured conformal radius in the original u coordinate is exactly

$$r_u = \frac{\pi}{2} \Im(\tau_0) |\vartheta_3(\tau_0)|^4.$$

Writing $r = e^{-\pi \Im \tau_0}$ and using only the $n = 1$ term of ϑ_3 explicitly gives

$$|\vartheta_3(\tau_0)| \leq |1 + 2re^{\pi i \Re \tau_0}| + 2 \sum_{n \geq 2} r^{n^2} < 0.9958.$$

Therefore

$$\boxed{r_u < \frac{\pi}{2} (0.847)(0.9958)^4 < 1.309.} \tag{5}$$

But

$$e^{117/400} = 1.3397727 \dots > 1.339.$$

Hence

$$\boxed{\log r_u < \frac{117}{400}.$$

Thus the level-10 route fails the arithmetic-holonomy entry condition even under the same maximally optimistic twenty-row penalty used below. This argument is elementary once the inverse- λ enclosure is known; no full potential-theory computation is required.

6 The $L(3, \chi_5)$ level-20 six-sheet realization

The six-sheet completion calculation reduces the fully elliptic-pure level-20 extension to the unique oldform line

$$\boxed{G_* = S + 4V_2S - 32V_4S}, \quad S = E_{4, \chi_5, 1},$$

with

$$L(G_*, 3) = -\frac{1}{2}L(3, \chi_5).$$

The finite elliptic Tate defects vanish, while the boundary matrix at the three remaining non-fillable cusp orbits has determinant $289/32 \neq 0$. Thus the exact conditional continuation surface is

$$X = X_0(20) \setminus \{Q, 3Q, 4Q\}, \quad z_0 = 2Q.$$

On the elliptic model

$$E: \quad Y^2 = m(m^2 - 2m + 5),$$

normalize

$$E(\mathbf{C}) \simeq \mathbf{C}/(\mathbf{Z} + \tau_E \mathbf{Z}), \quad \tau_E = \frac{1}{2} + 0.620969651446173011114040998188 \dots i.$$

Then

$$Q \leftrightarrow \frac{1}{6}, \quad 2Q \leftrightarrow \frac{1}{3}, \quad 3Q \leftrightarrow \frac{1}{2}, \quad 4Q \leftrightarrow \frac{2}{3}.$$

The real period is

$$\Omega_1 = 5.648750283918227598967579098086856 \dots,$$

and for the normalized arithmetic coordinate $x = (m - 1)/4 = q + O(q^2)$ one has

$$\left| \frac{dx}{dz}(1/3) \right| = \frac{\Omega_1}{2}, \quad r_x(X) = \frac{\Omega_1}{\rho_X(1/3)}.$$

The most optimistic possible twenty-row arithmetic profile has nineteen depth-zero rows and one depth-three extension row. Hence every one of the three nominal LCM columns is free on a 19-dimensional subspace and

$$\boxed{\tau_{\min} = 3 \left(1 - \frac{19^2}{20^2} \right) = \frac{117}{400}.} \tag{6}$$

Consequently the entry condition would require

$$\rho_X(1/3) < \Omega_1 e^{-117/400} = 4.2162003523116518836 \dots$$

6.1 Two-puncture monotonicity reduction

Fill the third bad cusp and put

$$Y_2 = E \setminus \left\{ \frac{1}{6}, \frac{1}{2} \right\}.$$

Then $X \subset Y_2$ and therefore $\rho_X \geq \rho_{Y_2}$. It is enough to construct on Y_2 a curvature- ≤ -1 metric whose density at $1/3$ exceeds the displayed threshold.

Let $b = \Im \tau_E$ and define the standard periodic Green coordinate

$$g_p(z) = -\log \left| \frac{\vartheta_1(z - p \mid \tau_E)}{\vartheta_1'(0 \mid \tau_E)} \right| + \frac{\pi \Im(z - p)^2}{b},$$

followed by the complete cusp model

$$u_p = g_p - \log g_p, \quad u_s = \frac{1}{2} \log \left(e^{2u_{1/6}} + e^{2u_{1/2}} \right).$$

Let

$$v_{20}(u, v) = \sum_{|k|, |\ell| \leq 20} c_{k, \ell} e^{2\pi i(ku + \ell v)}, \quad z = u + v\tau_E,$$

be the explicit real Fourier polynomial supplied with the certificate package, and set

$$\boxed{u_- = u_s + v_{20} - \frac{3}{40}, \quad \sigma = e^{u_-}.} \quad (7)$$

The required curvature inequality is

$$\Delta(u_s + v_{20}) \geq e^{-3/20} e^{2(u_s + v_{20})}.$$

Proposition 6.1 (Robust finite certificate). *For the explicit coefficient file used in (7), the periodic fundamental parallelogram admits a finite adaptive box cover on which the preceding curvature inequality is verified with conservative center-evaluation error budgets. The analytic cusp disks use radius 1.1×10^{-3} and have curvature-ratio lower bound*

$$0.8742197237325715 > e^{-3/20} = 0.8607079764250578 \dots$$

Outside those disks, a 1200×1200 structured cover followed by seven adaptive subdivisions terminates with no unresolved boxes. The successive unresolved counts are

$$48138, 51391, 23825, 24027, 30104, 32464, 2660, 0.$$

The computation remains successful after inflating every center evaluation by the explicit budgets

$$|\delta v|, |\delta u_s| \leq 10^{-7}, \quad |\delta \Delta v|, |\delta \Delta u_s| \leq 10^{-5}.$$

At the Taylor point a high-precision evaluation gives

$$\boxed{\sigma(1/3) = 4.267722328183018826861379346901214 \dots}$$

while

$$\Omega_1 e^{-117/400} = 4.216200352311651883607635379257822 \dots$$

Thus the base-point margin is $0.05152197587136 \dots$

Remark 6.2 (Certificate status). *The theta truncation after sixteen terms has a negligible analytic tail, and the finite Fourier/Taylor bounds and cusp estimates are explicit. The current implementation uses binary64 center evaluations together with error budgets vastly larger than the observed errors. A directed-rounding Arb/MPFI rerun would make the certificate maximally portable; the robust cover above is the programme's present computer-assisted certificate.*

Corollary 6.3 (Level-20 entry no-go). *Subject to the finite certificate of Proposition 6.1, the level-20 all-elliptic-pure six-sheet realization satisfies*

$$\boxed{\log r_x(X) < \frac{117}{400}.$$

Hence it fails the entry hypothesis of the CDT arithmetic-holonomy theorem even under the maximally favorable twenty-row denominator profile. A full CDT energy computation cannot rescue this realization.

6.2 Conceptual consequence

The example isolates a useful new Layer-C phenomenon. Exact purification in Layer B can enlarge the analytic leaf dramatically and still fail quantitatively because the sharp arithmetic lattice is too expensive. Thus the three layers support not only construction and classification theorems but structural no-go results:

$$\boxed{\text{extension purity} \not\Rightarrow \text{Diophantine viability.}}$$

The monotone entry obstruction provides an inexpensive first test before one invests in the full potential-theoretic energy of a candidate.

7 Global three-layer functoriality

The examples now force a distinction between three notions that were often conflated in recurrence-first treatments: transport of the homogeneous local system, transport of the mixed Eisenstein extension, and transport of a chosen Apéry realization. The following theorem packages the functorial statements used throughout the programme.

Theorem 7.1 (Three-layer Hecke–isogeny functoriality). *Let K be a characteristic-zero field and let Y_i be smooth modular/algebraic curves carrying elliptic families $\pi_i : E_i \rightarrow Y_i$. Fix an integer $r \geq 2$ and character data χ . After passage to a common finite algebraic cover Y' , suppose:*

(H1) *the elliptic families are related by an isogeny $\varphi : E_1 \rightarrow E_2$ of degree N ;*

(H2) *the resolved pure systems are*

$$\tilde{\mathbb{V}}_i = \text{Sym}^{r-1} H_{\text{dR}}^1(E_i/Y') \otimes L_\chi;$$

(H3) *distinguished period sections and local coordinates satisfy*

$$F_1 = hF_2, \quad t_1 = R(t_2),$$

with $h, R \in K(Y')$ and h generically nonzero;

(H4) *the relevant Eisenstein sources satisfy, on the common q -expansion sector,*

$$\Phi_1 = C\Phi_2, \quad C = \sum_d c_d V_d;$$

(H5) *orientation and boundary decompositions have been chosen compatibly with the correspondence.*

Then the following statements hold.

(A) Pure/isogeny transport. *The pullback φ^* is a horizontal isomorphism*

$$H_{\text{dR}}^1(E_2/Y') \xrightarrow{\sim} H_{\text{dR}}^1(E_1/Y'),$$

with inverse $N^{-1}\hat{\varphi}^$, and hence induces a horizontal isomorphism on every symmetric power. If a rational quotient coordinate is obtained by a finite map $\tilde{f} : \tilde{Y} \rightarrow Y$ of generic degree d , the descended pure system is $f_*\tilde{\mathbb{V}}$ and has generic rank $d \text{ rank } \tilde{\mathbb{V}}$.*

(B) Differential pullback–twist. *In rational first-order bases, if $DY_i = A_i Y_i$ and $Y_1 = UY_2$, then*

$$\boxed{A_1 U - DU - U A_2 = 0.}$$

When the coordinates differ by $t_1 = R(t_2)$ the connection is additionally rescaled by the chain-rule logarithmic derivative. This is differential pullback-twist equivalence; it does not by itself imply a same-index recurrence/UMAPS coboundary.

(C) Transport versus recanonicalization of an exact source. If both presentations lie on the rigid exact locus $\Phi_i = F_i Dt_i$, then

$$\boxed{\Phi_1 = hR'(t_2)\Phi_2.} \quad (8)$$

Thus functorial pullback of an extension and re-choosing the canonical exact source in a nonlinear coordinate differ by the ramification factor R' . Zeros or poles of R' can create apparent principal parts without changing the transported Layer-A motive.

(D) Eichler/Hecke transport. Define

$$C^{[j]} = \sum_d c_d d^{-j} V_d.$$

Then on zero-constant sectors

$$\boxed{D^{-j}C = C^{[j]}D^{-j}},$$

and in particular for the depth- r primitive

$$\Theta_1 = C^{[r]}\Theta_2.$$

If $P_C(s) = \sum_d c_d d^{-s}$, then

$$\boxed{L(\Phi_1, s) = P_C(s)L(\Phi_2, s)}, \quad L(\Phi_1, r) = P_C(r)L(\Phi_2, r).$$

Hence an isogeny/degeneracy correspondence acts diagonally on Mellin critical coordinates while the integrated correspondence records the extension transport.

(E) Critical-orientation-boundary functoriality. Let

$$\Psi = (\text{Crit}_{\text{or}}, \text{Res})$$

be the boundary-completed Layer-B coordinate map. On each resolved critical coordinate j , C acts by the scalar $P_C(j)$. On cusp constants it acts by the finite matrix obtained by transporting oldform constant terms to the target cusp orbits. Consequently the three kernels

$$K_{\partial} \subseteq K_{\text{or}} \subseteq K_{\text{sc}}$$

are functorial under correspondences preserving the chosen orientation decomposition, and target-zero/critical-invisible translations remain affine directions until removed by the finer boundary map.

(F) Companion descent. If $a : \mathbb{V} \rightarrow \mathbb{L}$ is a companion morphism and $a_*[\nu] = 0$, then horizontal transport of a and $[\nu]$ across the correspondence again gives a descended companion. On the exact scalar locus,

$$B_1 = F_1\Theta_1 = hF_2C^{[r]}\Theta_2.$$

Thus the companion kernel is functorial, but its dimension can change after direct image, symmetry projection, or change of target representation.

(G) Marked de Rham sections and coefficients. A change of cyclic/de Rham section inside a fixed horizontal local system can change the scalar Picard-Fuchs equation and recurrence without changing Layers A or B. If

$$F_1(t) = h(t)F_2(R(t)),$$

then

$$[t^n]F_1 = \sum_{k \geq 0} [t^n](h(t)R(t)^k)[u^k]F_2(u).$$

For $R(0) = 0$ this is finite; for $R(t) = \lambda t + O(t^2)$ it is triangular. Nonlinear coefficient transport is therefore weaker than same-index recurrence equivalence.

(H) Adelic Layer-C transport. A fixed finite-index change of stable integral lattice changes denominators only by a bounded factor and hence preserves exponential LCM entropy. A proved free-depth flag $U^\bullet$ and horizontal-integration profile e may be transported only when their all- n divisibility/integration hypotheses are preserved by the chosen realization. They are therefore Layer-C invariants, not invariants of the pure isogeny class alone. Likewise the complex holonomy leaf and monotone entry obstruction depend on the chosen coordinate/direct-image realization.

(I) Almost-everywhere homogeneous Frobenius invariance. At every good auxiliary prime $\ell \nmid N$, the rational ℓ -adic/crystalline homogeneous local systems of isogenous elliptic hosts, and hence their symmetric powers, have the same Frobenius characteristic polynomials after pullback. The extension Frobenius coordinate κ_ℓ and the free-depth lattice can nevertheless depend on the Layer-B extension and Layer-C realization.

Proof. (A) is contravariant functoriality of relative de Rham cohomology together with the dual-isogeny identity $\widehat{\varphi} \circ \varphi = [N]$; symmetric powers are tensor-functorial, and the direct-image rank statement follows fibrewise on the unramified locus. (B) follows by differentiating $Y_1 = UY_2$. For (C),

$$\Phi_1 = F_1 Dt_1 = hF_2 R'(t_2) Dt_2 = hR'(t_2) \Phi_2.$$

For (D), $DV_d = dV_d D$, hence $D^{-j}V_d = d^{-j}V_d D^{-j}$; Mellin transformation gives $L(V_d f, s) = d^{-s}L(f, s)$. Part (E) is the same diagonal action on resolved critical coordinates together with linear functoriality of cusp constant terms under degeneracy maps. Part (F) is naturality of extension pushforward under horizontal maps, with the displayed exact formula following from (D). Part (G) is coefficient extraction after algebraic substitution and records explicitly why a marked cyclic section is realization data. The bounded-lattice statement in (H) is elementary, while the remaining qualifications are hypotheses rather than automatic invariance claims. Finally (I) is the standard good-prime invariance of rational Tate/crystalline modules under isogeny. No claim about the explicit transport law of κ_ℓ is used. $\square$

Corollary 7.2 (Classification groupoid). *The natural equivalence problem is not the classification of recurrences but the groupoid of triples*

$$(\mathbb{V}, [\nu], \mathcal{R}_{\text{ad}})$$

under finite pullback/direct image, horizontal isogeny transport, Hecke/degeneracy action, target-zero translation, normalizer/orientation projection, and marked de Rham-section change. The three levels of equivalence are strictly different:

$$\text{same pure skeleton} \not\approx \text{same extension} \not\approx \text{same Apéry realization.}$$

Remark 7.3 (Finite-place frontier). *The theorem intentionally stops short of the conjectural identity*

$$\kappa_p(C\nu) \stackrel{?}{=} P_C(r) \kappa_p(\nu)$$

with the correct local Euler factors and character normalizations. Proving this statement for syntomic Eisenstein classes is now the principal missing functoriality theorem in the adelic theory.

8 Finite-place extension functoriality

The adelic theory should distinguish the full Frobenius matrix of the homogeneous local system from the finite-place coordinate of the mixed extension. The latter is naturally defined only modulo Frobenius coboundary, exactly as a complex normal function is defined modulo a change of splitting.

Let p be a good ordinary auxiliary prime. Syntomic Eisenstein classes are obtained by restricting the syntomic elliptic polylogarithm to torsion sections; Bannai–Kings and Sprang describe these classes on the ordinary locus in terms of p -adic Eisenstein–Kronecker/Katz-measure data [4, 5]. The syntomic logarithm sheaves are extensions in the exact category of filtered overconvergent F -isocrystals, so an Eisenstein class may be represented locally by an extension

$$0 \longrightarrow \mathbb{V}_p \longrightarrow \mathbb{E}_{\nu,p} \longrightarrow \mathbf{1} \longrightarrow 0.$$

Definition 8.1 (Frobenius-coboundary extension coordinate). *Choose a local splitting of the underlying vector bundle and write the Frobenius matrix as*

$$\Phi_{\mathbb{E}} = \begin{pmatrix} \Phi_{\mathbb{V}} & \kappa_p \\ 0 & 1 \end{pmatrix}. \quad (9)$$

The finite-place extension coordinate is the class

$$\boxed{[\kappa_p(\nu)] \in \operatorname{coker}(\Phi_{\mathbb{V}} - 1) = \mathbb{V}_p / (\Phi_{\mathbb{V}} - 1)\mathbb{V}_p}. \quad (10)$$

Proposition 8.2 (Splitting invariance). *The class (10) is independent of the chosen local splitting.*

Proof. If the chosen lift of $1 \in \mathbf{1}$ is replaced by $e_0 + v$, then

$$\Phi_{\mathbb{E}}(e_0 + v) = e_0 + \kappa_p + \Phi_{\mathbb{V}}v = (e_0 + v) + \kappa_p + (\Phi_{\mathbb{V}} - 1)v.$$

Hence κ_p changes by a Frobenius coboundary and its class in the cokernel is unchanged. $\square$

The same calculation shows why κ_p itself should not be treated as a canonical scalar before a Frobenius eigencoordinate or quotient has been selected.

Theorem 8.3 (Abstract finite-place Hecke–Eisenstein functoriality). *Let C be a finite algebraic Hecke/isogeny correspondence, of degree prime to the good auxiliary prime p , acting on a motivic Eisenstein/polylogarithmic extension ν . Assume the correspondence extends over the chosen good ordinary syntomic datum. Then:*

(i) *syntomic realization is functorial:*

$$r_{\operatorname{syn},p}(C\nu) = C r_{\operatorname{syn},p}(\nu);$$

(ii) *C commutes with the homogeneous Frobenius on the transported isocrystal and induces a map on the Frobenius-coboundary quotient;*

(iii) *the canonical finite-place extension coordinate satisfies*

$$\boxed{[\kappa_p(C\nu)] = C [\kappa_p(\nu)]} \quad \text{in } \operatorname{coker}(\Phi_{\mathbb{V}} - 1). \quad (11)$$

If q_p is a one-dimensional quotient of this cokernel on which C acts by the scalar λ_C , then

$$\boxed{q_p[\kappa_p(C\nu)] = \lambda_C q_p[\kappa_p(\nu)]}. \quad (12)$$

Proof. Polylogarithmic Eisenstein classes satisfy motivic/cohomological distribution relations under isogeny/Hecke correspondences [6]. Syntomic realization is a functorial realization of these classes in filtered overconvergent F -isocrystals; hence the distribution relation persists after syntomic realization. At a good prime the algebraic correspondence intertwines the Frobenius structures, so it descends to the cokernel of $\Phi_{\mathbb{V}} - 1$. Applying the correspondence to a triangular representative (9) gives (11); Proposition 8.2 makes the statement independent of all local splitting choices. Projection to a one-dimensional C -eigenquotient gives (12). $\square$

Corollary 8.4 (What remains to obtain the scalar $P_C(r)$ law). *Suppose a canonical one-dimensional finite-place regulator quotient $q_{p,r,\chi}$ is constructed for the Eisenstein extension attached to $L(r, \chi)$ and suppose the degeneracy operator V_d acts on this quotient by d^{-r} . Then for*

$$C = \sum_d c_d V_d, \quad P_C(s) = \sum_d c_d d^{-s},$$

one has

$$\boxed{q_{p,r,\chi}[\kappa_p(C\nu)] = P_C(r) q_{p,r,\chi}[\kappa_p(\nu)]}. \quad (13)$$

Proof. On the stated quotient, C acts by $\sum_d c_d d^{-r} = P_C(r)$, so apply Theorem 8.3. $\square$

Problem 8.5 (Dirichlet syntomic regulator identification). *Construct $q_{p,r,\chi}$ canonically, including character orientation and local Euler factors, and prove*

$$q_{p,r,\chi}[\kappa_p(\nu_{r,\chi})] = \mathcal{E}_p(r, \chi) L_p(r, \chi)$$

with an explicit normalization factor $\mathcal{E}_p(r, \chi)$. Together with Corollary 8.4, this would give the exact finite-place analogue of the complex Mellin identity

$$L(C\Phi, r) = P_C(r) L(\Phi, r).$$

Remark 8.6 (Status after the synthesis). *The correspondence law (11) is formal once one works with the canonical Frobenius-coboundary class and uses functoriality/distribution of syntomic Eisenstein classes. The genuinely open arithmetic content is now concentrated in Problem 8.5: selecting the canonical Dirichlet regulator quotient and identifying its scalar coordinate with the normalized p -adic Dirichlet L -value.*

9 Consequences and next calculation

The two linear/geometric bottlenecks that had remained in Layers B and C have therefore changed status. The Layer-B critical-orientation-boundary filtration is now an exact finite-dimensional theorem, and the free-depth part of the adelic Layer-C dictionary reproduces the arithmetic penalty of an independently successful irrationality proof exactly. The level-20 $L(3, \chi_5)$ six-sheet route is now reduced further: the monotone entry obstruction supplies a robust computer-assisted no-go before the full energy stage. The next structural frontier is therefore to design a genuinely different $L(3, \chi_5)$ realization with either a cheaper free-depth flag or a larger continuation leaf, and to formulate the finite-place Hecke–Eisenstein functoriality theorem for the extension coordinate κ_p .

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